

Networking Consulting Proposal

Team 9



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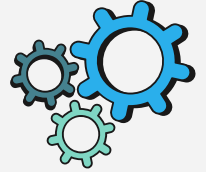


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Network Design Overview

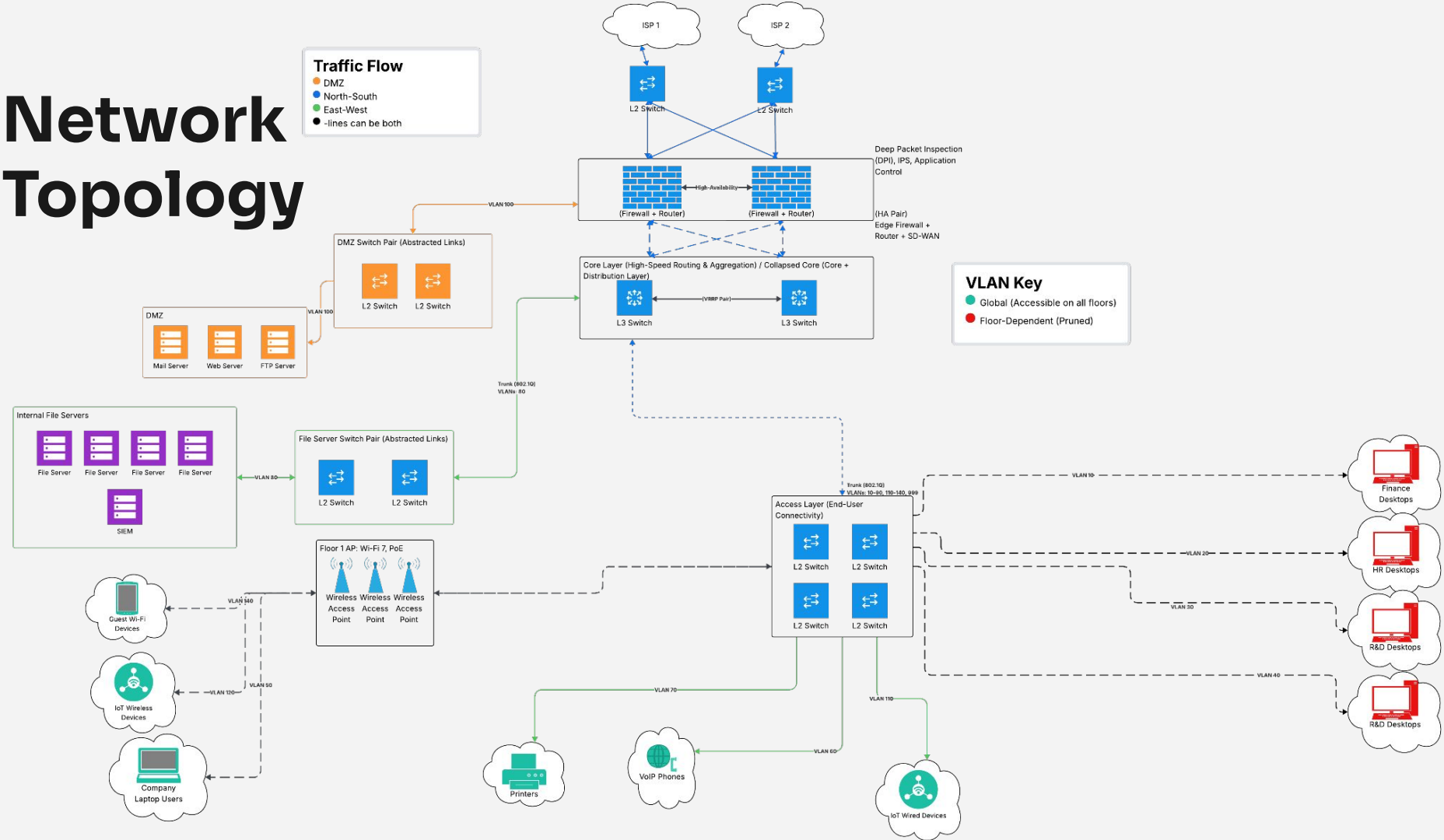


- ❖ Design a secure and scalable enterprise network
- ❖ Implement an all-in-one edge solution using the FortiGate 100F
- ❖ Segment the network using VLANs for security and performance
 - ❖ Support both internal (east-west) and external (north-south) traffic
- ❖ Ensure high availability using redundancy (HA firewall + VRRP)

Network Topology

Traffic Flow

- DMZ
- North-South
- East-West
- lines can be both



Network Architecture

- ❖ **Edge Layer:** FortiGate 100F (firewall, routing, DPI)
- ❖ **Core Layer:** Layer 3 switches (collapsed core with VRRP)
- ❖ **Access Layer:** Layer 2 switches connecting end devices
- ❖ **DMZ:** Isolated network for public-facing servers

Hardware Selection



- **FortiGate 100F:** all-in-one firewall, router, and DPI
- **FortiGate Layer 3 Switches:** inter-VLAN routing and core connectivity
- **FortiGate Layer 2 Switches:** connect end-user devices at the access layer
- **Wireless Access Points:** provide WiFi for users and guests
- Designed for **performance, security, and scalability**

Pros to FortiGate 100F

All-in-one security:

- Firewall, VPN, IPS, Web Filtering

Replaces multiple devices → simpler network

High performance (multi-Gbps, ASIC hardware)

Cost-effective vs competitors

Built-in SD-WAN:

- Dual ISP failover
- Load balancing
- Remote access support

Centralized management:

- Easy GUI
- Logging & analytics



Cons to FortiGate 100F

Steep learning curve

- Advanced features require networking knowledge

Complex licensing

- Multiple tiers (UTP, Enterprise, add-ons)

Ongoing maintenance required

- Regular firmware updates and patching needed for security

Hardware scalability limits

- Must upgrade device as the company grows

Potential stability issues

- Firmware bugs and occasional performance issues reported

VLAN & Network Segmentation

Network segmented by departments and device types
Created 14 VLANs to isolate traffic and improve security

Department VLANs:

- Finance, HR, R&D, Creative

Functional VLANs:

- Wireless, Voice, Printers, Servers, Management

Security-focused VLANs:

- DMZ, IoT (wired & wireless), VPN, Guest

Reduces broadcast domains and improves performance
Enables controlled communication between network segments



VLAN Structure Overview

VLAN ID	VLAN Name	Purpose
10	Finance	Financial systems and users
20	HR	Employee records and HR systems
30	R&D	Research and development systems
40	Creative	Design and media workstations
50	Wireless	Corporate laptops (Wi-Fi users)
60	Voice	VoIP phones
80	Server Farm	Internal servers and services
100	DMZ	Public-facing web, mail, FTP
110/120	IoT	Wired and wireless IoT devices
130	VPN	Remote user access
140	Guest	Visitor internet access

Subnetting Strategy

- **Base network:** 10.20.0.0/22
 - Provides 1022 usable IP addresses
- Used Variable Length Subnet Masking (**VLSM**)
- Subnets sized based on actual device requirements
- Large networks (IoT) → /25 subnets (126 usable)
- Medium networks (departments, wireless, VPN) → /26 subnets (62 usable)
- Small networks (servers, printers, DMZ) → /27–/28 subnets
- Minimizes wasted IP space while allowing scalability

Subnet Calculations

Corporate Wireless VLAN

- 40 devices
- $40 + 2 = 42$ required addresses
- Next power of 2 = 64
- → /26 subnet (62 usable IPs)

Wired IoT VLAN

- 110 devices
- $110 + 2 = 112$ required addresses
- Next power of 2 = 128
- → /25 subnet (126 usable IPs)

VLAN & Subnet Integration

- Each VLAN mapped to a unique subnet
- Default gateways configured on Layer 3 core switch
- Inter-VLAN routing handled at core layer
- DHCP assigns IPs per VLAN using centralized server
- DHCP relay (IP helper) used for cross-VLAN communication
- ACLs enforce security policies between VLANs

Traffic Flow (North-South & East-West)

- ❖ **North-South Traffic:** Communication between internal network and the internet
 - ❖ **East-West Traffic:** Internal communication between VLANs
- ❖ All external traffic passes through the FortiGate 100F for security inspection
- ❖ Deep Packet Inspection (DPI) is applied to detect threats and control applications
- ❖ DMZ handles inbound public traffic without exposing internal systems

Core Services & DMZ

DNS & DHCP: centrally managed for efficient IP addressing and name resolution

SIEM Server:

- Detectors:
 - **Ex:** "Alert" : User logs in from remote location in New York, same user logs in from California 5 minutes later.
- **Managed Devices:**
 - Firewall
 - Switches & routers
 - Mail, Web, File servers
 - DNS & DHCP queries

DMZ: hosts public-facing services (Web, Mail, FTP)

Firewall rules:

- Isolate DMZ from internal network
- Prevents external attacks from reaching sensitive systems
 - **Ex:** Deny - All data flow from DMZ to Finance VLAN
 - **Ex:** Limited access - HTTP request to update server from DMZ

Network Security

Zero Trust Remote Access: MFA + least privilege access for remote users

- Authentication Flow:
 1. User requests access
 2. Request Intercepted by Firewall
 3. Allow or Deny request
 4. Device Posture Check (Ensures device is secure)
 5. Require MFA
 6. Require SSO
 7. User is granted access to Least Privilege Account

IoT Isolation: IoT devices have a dedicated VLAN and restricted through the Firewalls

- Ex: Deny: Google Nest Thermostat attempts to access Finance VLAN
- Dedicated IoT VLANs:
 - VLAN 110: Wired IoT Devices
 - VLAN 120: Wireless IoT Devices

Wireless and Port Security:

- Port Restriction: Drop packets from unauthorized MAC addresses and send logs to SIEM
 - Ports are filtered by Firewall
- Wireless: WPA3 Enterprise

Business Continuity

High Availability:

- **Replicated devices:**

- Fortigate 100f: Running Fortigate Clustering Protocol (active-active)
 - Provides Load Balancing characteristics
- File server switches: Running LACP
 - Provides Load Balancing characteristics

Backup Power:

- Dual PSU for internal servers and core devices (Fortigate, switches):
 - **No hard Shutdowns:** System initiates graceful shutdown when PSU reports low battery.

Data Protection and Recovery:

- Hybrid Backups: Assess Recovery Point Objective (RPO)
 - Regular Local Backups (Hourly)
 - Cloud Backups (Weekly)
 - Provides easy disaster recovery

Conclusion

- ❖ Designed a secure and scalable enterprise network
 - ❖ Implemented VLAN segmentation and traffic control
 - ❖ Used FortiGate 100F for centralized security and DPI
- ❖ Ensured high availability with HA firewall and VRRP
- ❖ Protected internal systems with DMZ and Zero Trust model



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Thank you
Any questions?

